

Probability Theory

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These are my notes on probability theory, which I am studying for C271A and as an extension of what I learned in analysis. I hope to cover the relevant measure theory whenever possible, and I will also add completed practice problems as well.

Textbooks Used to Develop these Notes:

1. *Probability: Theory and Examples* - Rick Durrett
2. *Measure, Integration, and Real Analysis* - Sheldon Axler

Other Textbooks Referenced:

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Probability Spaces

We begin with some basic measure theory which will motivate the formulation of the σ -algebra and allow us to go from generality to specificity in defining probability spaces.

We begin by defining a probability space as a triple $(\Omega, \mathcal{F}, P)$. The first element Ω is called the set of outcomes or the "sample space", which is just an underlying set that can be defined as we wish.

Definition 1.1 (σ -algebra). *The second element $\mathcal{F}$ is called a σ -algebra, satisfying the following properties:*

1. $\Omega \in \mathcal{F}$
2. $\forall S \in \mathcal{F} : S \subseteq \Omega \wedge \Omega \setminus S \in \mathcal{F}$
3. *The countable (finite or countably infinite) union of sets in $\mathcal{F}$ is in $\mathcal{F}$:*

$$\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

We call the set S an event (specifically an event with an assigned probability, which we shall see later).

Lemma 1.1. *We can derive other properties of sets in the σ -algebra, such as:*

1. $\emptyset \in \mathcal{F}$
2. $\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:

$$\bigcap_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

3. $\forall S, T \in \mathcal{F} : S \setminus T \in \mathcal{F}$

Proof. For $\emptyset \in \mathcal{F}$, this follows from the fact that $\Omega \in \mathcal{F}$ and the complement of Ω then must be in $\mathcal{F}$.

The second statement follows from DeMorgan's Laws. If $S_n \in \mathcal{F}$, then $S_n^c \in \mathcal{F}$, which tells us that:

$$\bigcup_{n \in \mathbb{N}} S_n^c = \left(\bigcap_{n \in \mathbb{N}} S_n \right)^c \in \mathcal{F}$$

Once again by the complement axiom, we see that:

$$\bigcap_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

As for the third statement, we know that $S \setminus T = S \cap (\Omega \setminus T)$. Since $T \in \mathcal{F}$, then $\Omega \setminus T \in \mathcal{F}$, so by statement 2, we have that statement 3 follows. $\square$

Lemma 1.2. *The intersection of a family of σ -algebras for a set Ω is a σ -algebra.*

Proof. Let us call the family $\{\mathcal{F}_\alpha : \alpha \in I\}$ such that we have valid measurable spaces $(\Omega, \mathcal{F}_\alpha)$, then:

$$\mathcal{F} := \bigcap_{\alpha \in I} \mathcal{F}_\alpha$$

is a valid sigma algebra such that $(\Omega, \mathcal{F})$ is a measurable space. We see that:

$$\forall \alpha \in I : \Omega \in \mathcal{F}_\alpha \implies \Omega \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha$$

so property 1 in Definition 1.1 is fulfilled. Now, for some S in the intersection:

$$\begin{aligned} S \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha &\iff \forall \alpha \in I : S \in \mathcal{F}_\alpha \\ &\implies \forall \alpha \in I : S^c \in \mathcal{F}_\alpha \iff S^c \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha \end{aligned}$$

thus property 2 is fulfilled. Now consider some $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$, then:

$$\begin{aligned} \forall n \in \mathbb{N} : S_n \in \mathcal{F} &\implies \forall \alpha \in I : S_n \in \mathcal{F}_\alpha \\ \implies \forall \alpha \in I : \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}_\alpha^{\mathbb{N}} &\implies \forall \alpha \in I : \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F}_\alpha \\ &\implies \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F} \end{aligned}$$

so we have verified property 3. This is quite similar to topologies, where the intersection of topologies is a topology, but the union is not necessarily a topology. $\square$

Consider some sample space $\Omega := \{1, 2, 3, 4\}$, then consider:

$$\mathcal{F}_1 := \{\Omega, \emptyset, \{1\}, \{2, 3, 4\}\}$$

$$\mathcal{F}_2 := \{\Omega, \emptyset, \{2\}, \{1, 3, 4\}\}$$

both of which are σ -algebras. If we take the intersection:

$$\mathcal{F}_3 := \mathcal{F}_1 \cap \mathcal{F}_2 = \{\Omega, \emptyset\}$$

which is the "coarsest" or trivial σ -algebra. Now taking the union:

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\Omega, \emptyset, \{1\}, \{2\}, \{2, 3, 4\}, \{1, 3, 4\}\}$$

but by Definition 1.1 property 3, we would need $\{1, 2\}$ to belong in the union, thus the union is not necessarily a σ -algebra.

1.1 Motivating the σ

The third and final element of the triple is P , which we call the probability measure. A measure is defined as a following:

Definition 1.2. $(X, \mathcal{F})$ is a measurable space on which we can put a measure $\mu : \mathcal{F} \rightarrow [0, \infty]$ with the properties:

1. For any interval i

To motivate the use of σ -algebras, we look at a simple probability based example. Consider a weighted coin where the conductor of the experiment is not aware of the weightage. Then:

$$\Omega := \{H, T\}$$

However, our σ -algebra would be:

$$\mathcal{F} := \{\Omega, \emptyset, \{H, T\}\}$$

because

we fix our attention to subsets of $\mathbb{R}$. We defined the notion of outer measure of subsets of $\mathbb{R}$ as the following:

$$\forall A \subseteq \mathbb{R} : |A| := \inf \left\{ \sum_{n \in \mathbb{N}} |I_n| : A \subseteq \bigcup_{n \in \mathbb{N}} I_n \right\}$$

where I_n is an open interval (or closed works as well) whose measure we defined as the difference between its endpoints $b - a$.